

BRAF fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes BRAF gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 15 fusion events. 15/15 fusions (100.0%) were in-frame. 9/15 fusions (60.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=1$, not statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 100.0% in-frame and 60.0% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 15 analyzed fusion events, identifying 8 distinct fusion partner genes. The most frequent partner was SND1, observed in 5 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 9/15 (60.0%) of in-frame fusions retained the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.015984.

Tables

domain_retention tables

frame_domain_contingency_table

9	0
6	0

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	15	9	0	6	0.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AP3B1	2
BCL2L11	1
FAM114A2	2
MACF1	2
MKRN1	1
SND1	5
SUGCT	1
ZC3HAV1	1

Per-event results (results.tsv)

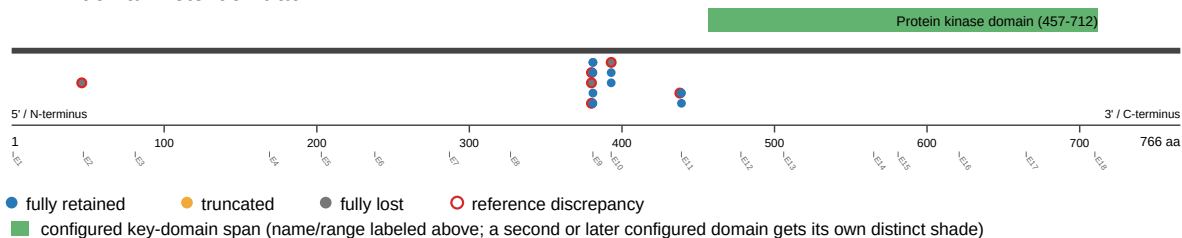
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3T0-01-1	TCGA-EL-A3T0-01		AP3B1-BRAF	AP3B1	in-frame	three_prime	140787584	9	381	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA AP3B1-BRAF Fusion	in-frame
EVT-TCGA-ET-A2MX-01-2	TCGA-ET-A2MX-01		BCL2L1-BRAF	BCL2L1	in-frame	three_prime	140783157	10	393	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA BCL2L1-BRAF Fusion	in-frame
EVT-TCGA-EL-A3T0-01-3	TCGA-EL-A3T0-01		BRAF-AP3B1	AP3B1	in-frame	five_prime	140794308	8	380	False	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-AP3B1 Fusion	in-frame
EVT-TCGA-ET-A3BN-01-4	TCGA-ET-A3BN-01		BRAF-FAM114A2	FAM114A2	in-frame	five_prime	140783021	10	438	False	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-FAM114A2 Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-5	TCGA-DE-A0Y2-01		BRAF-MACF1	MACF1	in-frame	five_prime	140794308	8	380	False	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-MACF1 Fusion	in-frame
EVT-TCGA-EL-A3ZK-01-6	TCGA-EL-A3ZK-01		BRAF-SND1	SND1	in-frame	five_prime	140787548	9	393	False	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-SND1 Fusion	in-frame
EVT-TCGA-KS-A41I-01-7	TCGA-KS-A41I-01		BRAF-SND1	SND1	in-frame	five_prime	140794308	8	380	False	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-SND1 Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-FY-A40N-01-8	TCGA-FY-A40N-01		BRAF-SUGCT	SUGCT	in-frame	five_prime	140924566	1	46	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		NA BRAF-SUGCT Fusion	in-frame
EVT-TCGA-ET-A3BN-01-9	TCGA-ET-A3BN-01		FAM114A2-BRAF	FAM114A2	in-frame	three_prime	140781693	11	439	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA FAM114A2-BRAF Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-10	TCGA-DE-A0Y2-01		MACF1-BRAF	MACF1	in-frame	three_prime	140787584	9	381	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA MACF1-BRAF Fusion	in-frame
EVT-TCGA-J8-A3O1-01-11	TCGA-J8-A3O1-01		MKRN1-BRAF	MKRN1	in-frame	three_prime	140787584	9	381	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA MKRN1-BRAF Fusion	in-frame
EVT-TCGA-DJ-A3UZ-01-12	TCGA-DJ-A3UZ-01		SND1-BRAF	SND1	in-frame	three_prime	140787584	9	381	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_junctions_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3ZK-01-13	TCGA-EL-A3ZK-01		SND1-BRAF	SND1	in-frame	threonine	140783157	10	393	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame
EVT-TCGA-FY-A40N-01-14	TCGA-FY-A40N-01		SND1-BRAF	SND1	in-frame	threonine	140787584	9	381	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame
EVT-TCGA-KS-A4ID-01-15	TCGA-KS-A4ID-01		ZC3HAV1-BRAF	ZC3HAV1	in-frame	threonine	140781693	11	439	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA ZC3HAV1-BRAF Fusion	in-frame

domain retention outliers

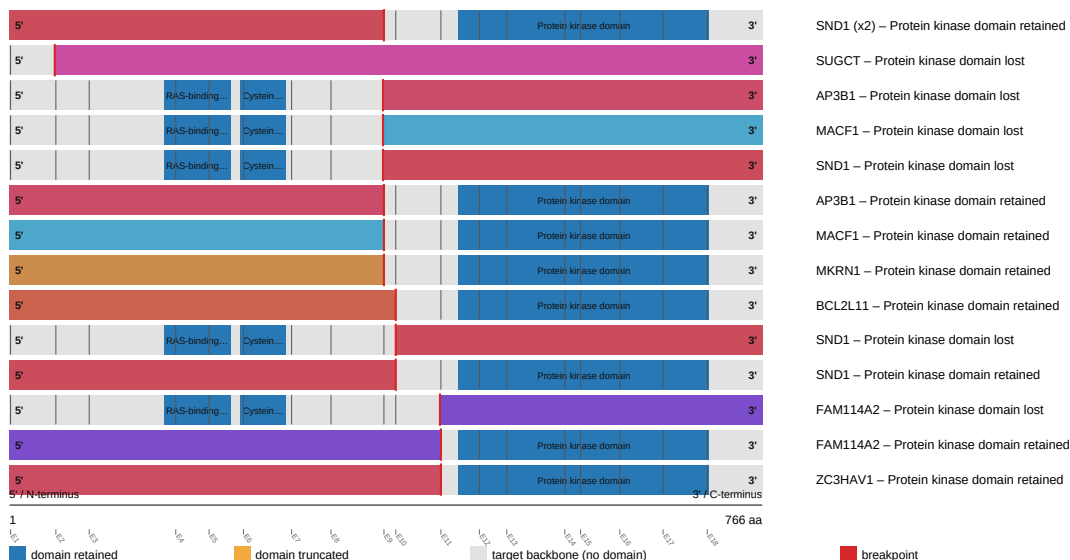
BRAF domain-retention track



fusion schematic

BRAF fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

reference comparison

Reference vs thca_tcga_pan_can_atlas_2018

